Fault Detection and Isolation - Report

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I. Introduction

This report covers the different implementations of fault detection and isolation methods for discrete-time systems. It shows and analyses the results obtained based on the theoretical explanations given by Prof. M. Kinnaert in the *Model-Based and Data-Driven Fault Detection and Isolation* course.

The discrete-time state space model is given by

$$\begin{cases} x[k+1] = Ax[k] + Bu[k] + E_d d[k] + E_f f[k] \\ y[k] = Cx[k] + Du[k] + G_d d[k] + G_f f[k] \end{cases} \tag{1}$$

where $x \in \mathbb{R}^{n\times 1}$ is the state, $u \in \mathbb{R}^{m\times 1}$ the input to the system, $y \in \mathbb{R}^{p\times 1}$ the output of the system, $d \in \mathbb{R}^{n_d\times 1}$ the disturbance and $f \in \mathbb{R}^{n_f\times 1}$ the fault. The exact values for those matrices are given in Appendix A.

II. Parity relations based method

A. Overview of the method

Based on the state space representation (1), $y[k]$ can be expressed as a function of the state at time instance $k-s$ and the input, disturbance and fault from time instance $k-s$ up to k. This was derived by recurrence and the resulting equation, in a matrix form, is

$$Y_s(k) = \Theta_s x(k-s) + T_s^u U_s(k) + T_s^d D_s(k) + T_s^f F_s(k). \tag{2}$$

The "*signal matrices*" are built as follows

$$Y_s(k) = \begin{bmatrix} y[k-s] \\ \vdots \\ y[k] \end{bmatrix} \quad \cdots \quad F_s(k) = \begin{bmatrix} f[k-s] \\ \vdots \\ f[k] \end{bmatrix}, \tag{3}$$

and the "*weight matrices*":

$$\Theta_s = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^s \end{bmatrix}, \quad T_s^u = \begin{bmatrix} D & \mathbf{0} & \cdots & \mathbf{0} \\ CB & D & \ddots & \vdots \\ \vdots & \ddots & \ddots & \mathbf{0} \\ CA^{s-1}B & \cdots & CB & D \end{bmatrix}$$
$$T_s^{d/f} = \begin{bmatrix} G_{d/f} & \mathbf{0} & \cdots & \mathbf{0} \\ CE_{d/f} & G_{d/f} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \mathbf{0} \\ CA^{s-1}E_{d/f} & \cdots & CE_{d/f} & G_{d/f} \end{bmatrix} \tag{4}$$

Because of their nature, disturbances are unknown (stochastic noise, nonlinearities, unmeasured input signal, ...) and the state is often not measured. To only keep from (2) the fault and the measured signals, they are split from the other terms and both sides of the equation is then multiplied by Ω_s.

The unknown side of the equation can then be cancelled with a clever choice for Ω_s, i.e. a basis of the left null space of $\begin{bmatrix}\Theta_s & T_s^d\end{bmatrix}$:

$$\Omega_s \left(Y_s(k) - T_s^u U_s(k) - T_s^f F_s(k)\right) = \mathbf{0}. \tag{5}$$

It is worth mentioning the existence condition for Ω_s, which depends on the rank of the matrix to cancel:

$$\text{rank}\begin{bmatrix}\Theta_s & T_s^d\end{bmatrix} < (s+1)p \tag{6}$$

This leads to the definition of the residual vector, which is equal to $\mathbf{0}$ in case there is no fault and is computed by

$$r[k] = \Omega_s \left(Y_s(k) - T_s^u U_s(k)\right). \tag{7}$$

B. Fault detection

Applied to a system with 1 input and 2 outputs, a simulation has been done with $s = 2$. It contains a sinusoidal disturbance and a fault on the sensors. Fig. 1 shows what is computed by the simulation in a case where no fault is present.

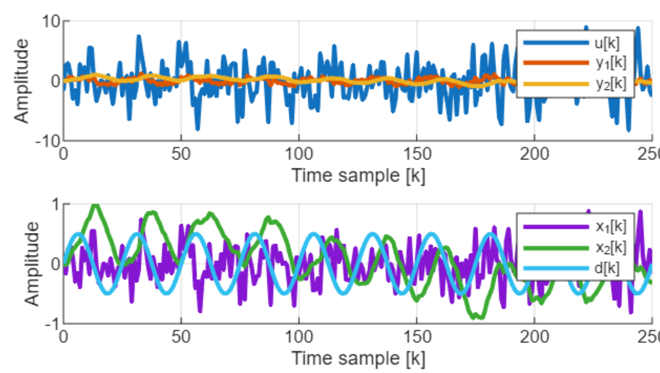

Fig. 1. Measured signals (top) and internal signals (bottom) in a faultless scenario.

The simulation is ran with a constant bias error on each of the two sensors and the resulting residual vector norm is shown on Fig. 2.

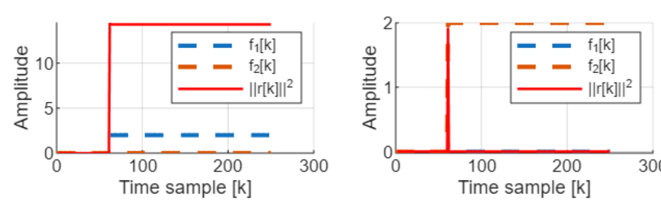

Fig. 2. Applied fault (on state 1 on the left plot and state 2 on the right plot) and resulting residual.

The residual on the right subplot indicates that the fault can be detected but is forgotten after a few samples. This is because the second state integrates the input, meaning a constant error will not be detected. On the other hand, the first state is simply amplifying the input. A constant difference between the expected output and the measured one is therefore easily detected.

For the same reason, a ramp-shaped fault will generate a ramp shaped-residual when applied on the first state and a constant residual in the other case, as shown on Fig. 3.

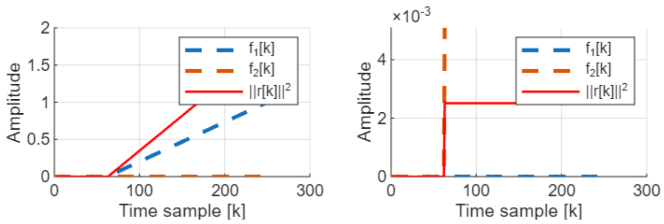

Fig. 3. Applied fault (on state 1 on the left plot and state 2 on the right plot) and resulting residual. The second fault looks like a step due to the scale difference between the residual vector norm and the fault signal.

C. Fault Isolation

To isolate the fault, i.e. determine which is the faulty sensor, residuals must be crafted by considering some faults as disturbances using a incidence matrix. Because on the existence condition on the left null space of $\begin{bmatrix}\Theta_s & T_s^d\end{bmatrix}$ given by (6), a residual generally cannot be built to control a single fault.

To understand how fault isolation works, the system is slightly modified by adding a second sensor on the first output. Three distinct residuals are then built, each targeting all but one fault according to the following incidence matrix:

	f_1	f_2	f_3
r_1	0	$\times$	$\times$
r_2	$\times$	0	$\times$
r_3	$\times$	$\times$	0

To generate r_1, the disturbance matrices E_d, G_d and the fault matrices E_f, G_f are modified. By moving some columns from the fault matrices to the disturbance matrices as follows,

$$\begin{cases} E_d = \begin{bmatrix}E_{d_1}\end{bmatrix} \\ E_f = \begin{bmatrix}E_{f_1} & E_{f_2} & E_{f_3}\end{bmatrix} \\ G_d = \begin{bmatrix}G_{d_1}\end{bmatrix} \\ G_f = \begin{bmatrix}G_{f_1} & G_{f_2} & G_{f_3}\end{bmatrix} \end{cases} \Rightarrow \begin{cases} E_d = \begin{bmatrix}E_{d_1} & E_{f_1}\end{bmatrix} \\ E_f = \begin{bmatrix}E_{f_2} & E_{f_3}\end{bmatrix} \\ G_d = \begin{bmatrix}G_{d_1} & G_{f_1}\end{bmatrix} \\ G_f = \begin{bmatrix}G_{f_2} & G_{f_3}\end{bmatrix} \end{cases} \tag{8}$$

the residual r_1 will now only detects faults on the 2nd and 3rd sensors. The same approach allows to generate 3 residuals, and therefore a fault can be isolated by inspecting which residual has not increased.

Fig. 4 shows a fault applied on different sensors and the corresponding residuals. Looking at the ones that are activated and combining it with the knowledge of the incidence matrix shows it is working as intended.

If multiple errors occur at the same time, there is no way of determining which sensor is faulty as shown on Fig. 5. As

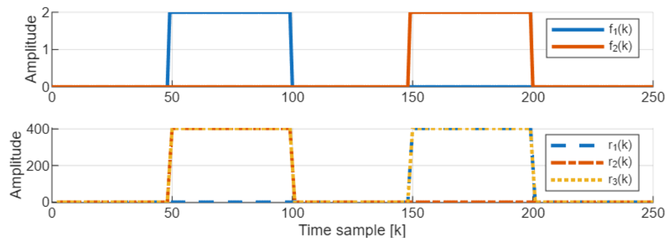

Fig. 4. Applied faults (top) and resulting residuals (bottom).

all the residuals are active between the 100th and the 150th samples, faults are still detected but cannot be isolated.

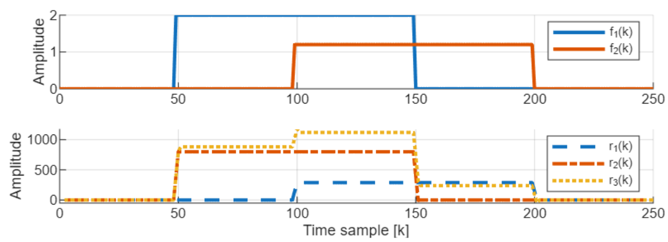

Fig. 5. Applied faults (top) and resulting residuals (bottom).

III. Observer Based Method

In this section, faults will be detected and isolated using methods based on state estimation. Depending on the type of model, Luenberger estimators or Kalman filters will be applied to build residuals.

A. Deterministic model

Starting from the discrete-time state space model (1), it is here considered to be free of any disturbance. A Luenberger observer is then built such that $\hat{x}$ estimates the true state x using

$$\hat{x}[k+1] = A\hat{x}[k] + Bu[k] + L\left(y[k] - C\hat{x}[k] - Du[k]\right). \tag{9}$$

The residual is then defined as the difference between the estimated output $\hat{y}$ and the measured one

$$\begin{aligned} &\Leftrightarrow r[k] = y[k] - \hat{y}[k] \\ &\Leftrightarrow r[k] = y[k] - C\hat{x}[k] - Du[k] \end{aligned} \tag{10}$$

To go towards fault isolation, the C and D matrices are truncated of one or more outputs, which build observers of a subsystem. This allows to detect faults that appear on a specific subset of sensors.

There is a condition on the existence of this observer which is quite obvious: $(A, C_{i,\cdot})$ should be observable, where $C_{i,\cdot}$ represents the i^{th} row of C[1].

The objective is to perform fault detection and isolation on the output signal of a tree-phased signal, which model is derived in Appendix B. A first set of observers were built using the following incidence matrix:

[1]This condition can be weakened as the observer could be built to only observe the observable subsystem.

	f_1	f_2	f_3
r_1	$\times$	0	0
r_2	0	$\times$	0
r_3	0	0	$\times$

Applied on the faulty signal, the measured signals and the residuals are shown on Fig. 6 and the bias on the second signal is clearly identified. The initial estimated state is randomly chosen, which explains why the residuals are not null in the first few samples.

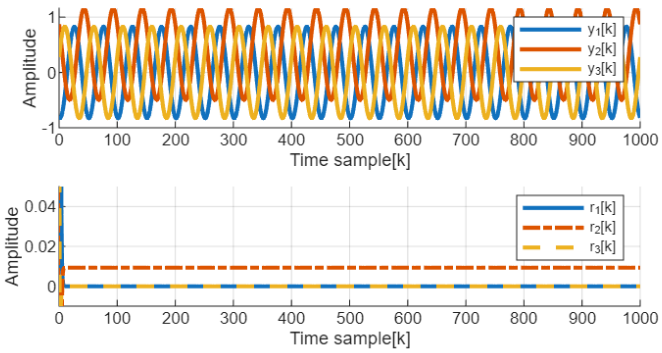

Fig. 6. Measured signals (top) and resulting residuals (bottom).

A different incidence matrix:

	f_1	f_2	f_3
r_1	0	$\times$	$\times$
r_2	$\times$	0	$\times$
r_3	$\times$	$\times$	0

was applied on the same signal and the residuals also allow to isolate the fault, as the 1st and the 3rd residuals are non-zero on Fig. 7.

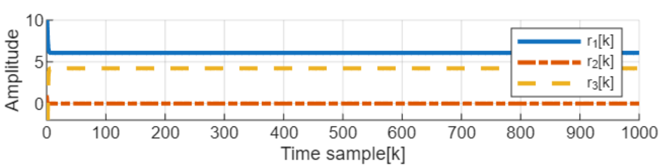

Fig. 7. Residuals with the alternative incidence matrix.

Appendix A
Exact Ship Model

The model used in the simulations is, in continuous time,

$$\dot{x}(t) = A_c x(t) + B_c u(t) + E_{d,c} d(t) + E_{f,c} f(t), \quad (11)$$

$$y(t) = C_c x(t) + D_c u(t) + G_{d,c} d(t) + G_{f,c} f(t), \quad (12)$$

with

$$A_c = \begin{bmatrix} b\eta_1 & 0 \\ 1 & 0 \end{bmatrix}, \quad B_c = \begin{bmatrix} b \\ 0 \end{bmatrix}, \quad E_{d,c} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad (13)$$

$$E_{f,c} = \mathbf{0}, \quad \text{and} \quad \begin{cases} b = 2.2 \\ \eta_1 = -10 \end{cases}. \quad (14)$$

The output equation matrices are

$$C_c = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad D_c = 0, \quad (15)$$

$$G_{d,c} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad G_{f,c} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (16)$$

The discrete-time model in (1) is finally obtained using a ZOH discretization with sampling time $T_s = 0.2$ seconds.

Appendix B
Three-Phased Signal Model

Fig. 8 shows the signal in a case no fault is applied. Knowing the sampling period $T_s = 1\text{ms}$, the frequency of the signal is computed by

$$\omega_{\text{exc}} = \frac{2\pi}{50T_s} \quad (17)$$

because there are 50 measurements per period.

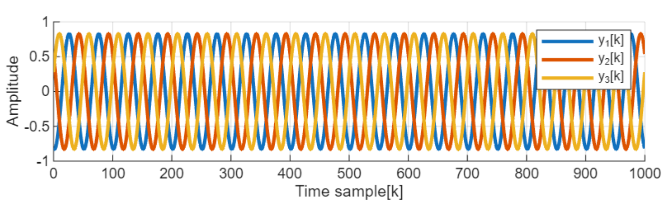

Fig. 8. Three-phased signal measured by 3 (faultless) sensors.

The continuous-time state space model of this system (which has no input) is given by

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & \omega_{\text{exc}} \\ -\omega_{\text{exc}} & 0 \end{bmatrix} \mathbf{x}. \quad (18)$$

Because there is an offset between the three phases, the output relationship is expressed as follows, where $\mathbf{f}$ is the fault vector:

$$\mathbf{y} = \begin{bmatrix} \cos(0) & \sin(0) \\ \cos(\frac{-2\pi}{3}) & \sin(\frac{-2\pi}{3}) \\ \cos(\frac{2\pi}{3}) & \sin(\frac{2\pi}{3}) \end{bmatrix} \mathbf{x} + \mathbf{f}. \quad (19)$$

This continuous-time model is then discretized using the ZOH with the known sampling frequency.